How the most successful innovators do it

BRING YOUR BREAKTHROUGH IDEAS TO LIFE

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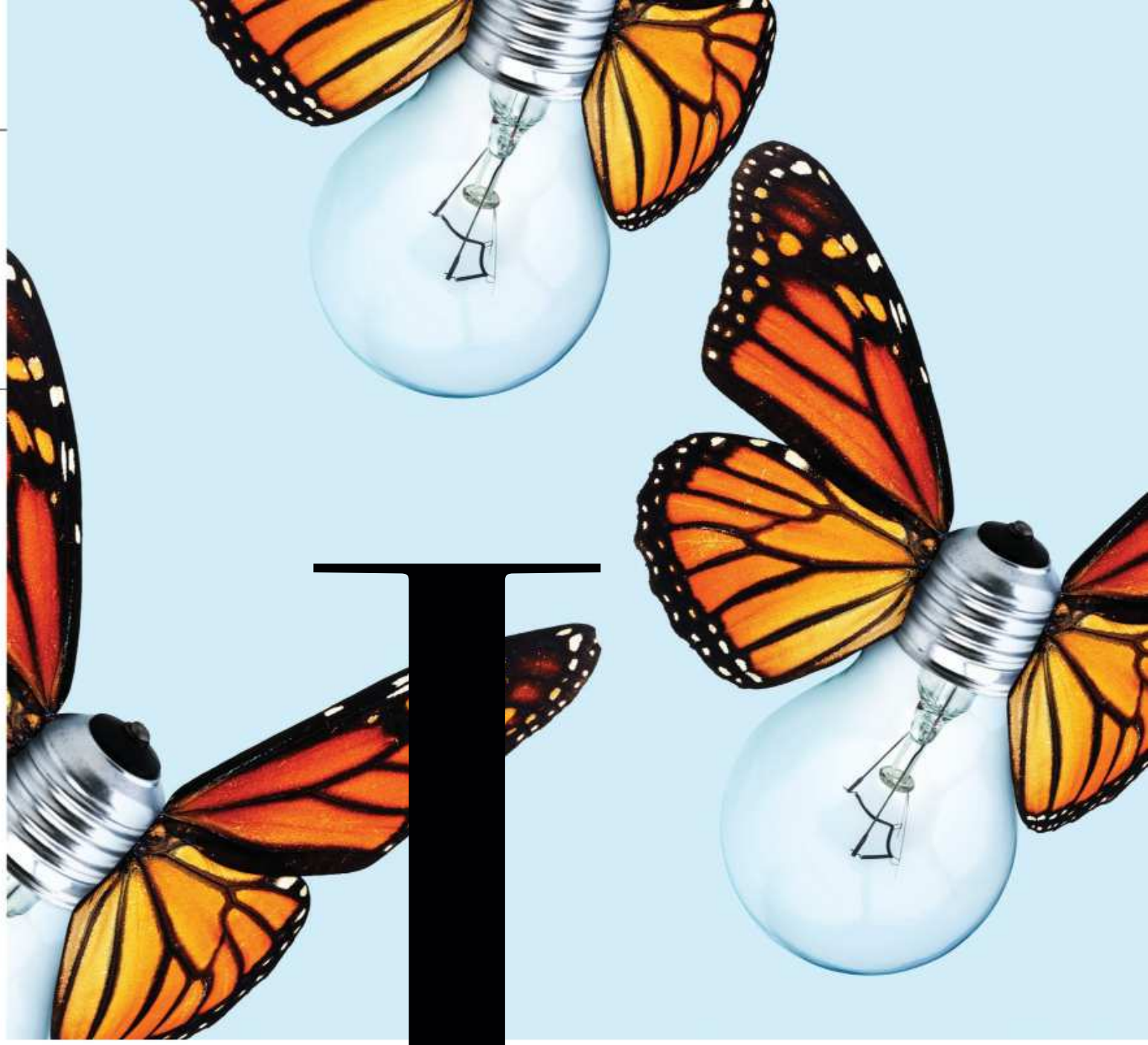

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Innovation

IDEA IN BRIEF

THE PROBLEM

Despite recent digital advances and an improved understanding of the innovation process, game-changing offerings remain hard to come by.

WHY IT HAPPENS

Existing innovation models are unrealistic and often incomplete, failing to incorporate the digital aspect of innovation and emphasizing speed over deep reflection. They gloss over pitfalls and biases and may focus so intensely on users that other key stakeholders are neglected.

THE REMEDY

Five practices can help bring breakthrough ideas to fruition: Attend closely and with fresh eyes, step back to gain perspective, imagine unorthodox combinations, experiment quickly and smartly, and navigate potentially hostile environments both inside and outside the company.

In 2003 the Indian environmental researcher Narayana Peesapaty spotted an alarming trend: Groundwater levels in the region of Hyderabad were falling precipitously. He examined rainfall records but found nothing to explain the drop. Looking deeper, he discovered that the culprit was a change in agricultural practices. Many area farmers had abandoned millet—a traditional crop increasingly regarded as "the poor man's food"—in favor of rice, a thirsty crop that requires 60 times as much water to grow. And because they had access to heavily subsidized electricity, the farmers were continuously pumping water into their fields.

Peesapaty tried to influence agricultural policies by documenting the problem in government reports, to no avail. So he looked instead for ways to boost demand for millet. He hit on the idea of turning it into "edible cutlery"—a solution that could attack not just the groundwater deficit but also the scourge of plastic waste. Peesapaty quit his job to pursue the project. A decade later, after a video he posted about the cutlery went viral, orders began pouring in. Two crowdfunding campaigns exceeded their targets by more than twelvefold, and the first corporate orders shipped in 2016. It's too soon to know whether groundwater levels have stabilized. But many farmers have already resumed growing the more sustainable crop, and to further boost production, the government declared 2018 the National Year of Millets.

As Peesapaty's story demonstrates, there are two potential routes to any solution: *conformity* (in this case, trying to use established channels to affect policy) and *originality*. The first is adequate for many everyday challenges. But for thornier problems, more-divergent thinking may be required.

As academics with a long-standing interest in attention, sense making, innovation, and digital transformation, we have spent the past decade researching pioneering thinkers and changemakers in a wide range of fields, from entrepreneurs to medics to chefs. Our work with corporate clients has included running top-team innovation workshops, leading full-scale acceleration programs, and orchestrating enormous transformation journeys. We have also interviewed and surveyed hundreds of executives involved in innovation efforts. Through these efforts we have identified recurrent patterns in the evolution of breakthrough ideas and constructed a five-part framework for developing them and ensuring their survival.

Unconventional thinkers focus their attention closely and with fresh eyes, step back to gain perspective, imagine unorthodox combinations, experiment quickly and smartly, and navigate potentially hostile environments outside and within their organizations. The challenge throughout is to overcome biases and mental models that may constrain creativity or doom a great idea.

In this article we'll describe the five elements of the framework and explore how digital tools can augment them. But first let's look at why game-changing innovation remains so difficult despite organizational and societal pressure for transformative results.

The Elusiveness of Breakthrough Innovations

The digital advances of the past two decades have enabled a much broader population than ever before to express creative intelligence. Unconventional thinkers the world over have unprecedented access to the distributed knowledge, talents, capital, and consumers they need to create a start-up or a movement around a great idea. Innovation has been thoroughly democratized.

And yet breakthrough offerings remain hard to come by. Apart from the transformation of services powered by mobile apps and the internet, we have not seen spectacular surges of innovation across sectors. The economists Tyler Cowen and Robert Gordon have spoken of innovation stagnation. The business thinker Gary Hamel notes that corporations are awash in ideas that fall into one of two buckets: incremental no-brainer or flaky no-hoper. And in our consulting work with innovation teams we see many promising ideas become superficial, narrow, or skewed—or perish altogether.

The lack of progress is surprising given that companies have an improved understanding of the innovation process,

driven in large part by design thinking and lean start-up methodologies. Terms such as "user centered," "ideation," and "pivot" have become commonplace and have changed the way business leaders think about creating new offerings. Yet for all this guidance, only 43% of corporations have what experts consider a well-defined process for innovation, according to the research firm CB Insights.

When we talk with entrepreneurs and executives about existing innovation frameworks, their criticisms center on three overlapping issues. First, the models are *unrealistic:* The still-influential waterfall, or stage-gate approach, for example, is overly linear, with little regard for the constant zigzagging between activities that may be called for. Elmar Mock, a serial entrepreneur who co-created the Swatch, put it this way in a 2016 podcast: "The very natural instinct for an innovator is to move in a nonlinear way, to go from concepts to know-how back to concept, to relook for new know-how, to change the concept again." Second, the models are *incomplete:* They don't incorporate the digital aspect of innovation or show how it relates to the "humancentric" principles enshrined in design thinking. They emphasize action and fast iteration (pillars of the lean start-up methodology), but in doing so they tend to downplay what Wharton's Adam Grant calls strategic procrastination—allowing yourself time for deep reflection. Third, the models are *misleading:* They gloss over the pitfalls and biases that may constrain creativity. And by focusing so intensely on users, they minimize the roles of other stakeholders and the need for inventiveness in mobilizing support to establish and deploy novel offerings.

Regarding this last point, executives recognize that to devise ingenious innovations, they must break paradigms and shift mindsets—but when it comes to delivery, they often lapse into standard ways of thinking. Consider the failed Sony Reader. All the creativity that went into its development was undone by a lack of originality in execution. Sony neglected to enlist the book publishing industry as an ally—a mistake Amazon did not make when it launched the technically inferior but hugely successful Kindle, 14 months later. To make your stellar innovation thrive, approach unconventional partners, identify underutilized channels, and invent new business models. Put as much creative energy into introducing and delivering offerings as you did into generating them. Sony engineered an elegant *device,* but Amazon designed an original *solution.*

Our framework complements design thinking, lean start-up, the business model canvas, and other innovation strategies. It is more accepting of the messiness inherent in developing a truly breakthrough solution, recognizing that the activities involved relate to one another in unpredictable, nonlinear ways. The elements of our framework are not unique, but collectively they capture the full scope and reality of the innovation process, including the critical role of reflection in conceiving opportunities and the level of organizational reinvention needed in the final push to market.

Let's turn now to those five elements.

Attention: Look Through a Fresh Lens

Attention is the act of focusing closely on a given context to understand its dynamics and latent needs. The trouble is that expertise often interferes, directing people's attention and unconsciously blinding them to radical insights. The French call this *déformation professionnelle:* the tendency to observe reality through the distorting lens of one's job or training. To combat that bias, question what perspective drives your attention and what you may be missing as a result.

Take the case of Billy Fischer, a U.S. infectious disease expert who regularly traveled to rural Guinea to fight the Ebola epidemic. In May 2014 he saw that the recommended approach was not working: The local treatment facility was focused on containing the spread by isolating anyone exposed to the virus, but people were hiding to avoid being quarantined. Talking with patients, Fischer realized that the problem was fear: The mortality rate for patients in quarantine was 90%—so people understandably saw it as a death sentence. He insisted that the clinic prioritize patient recovery instead. Testing new treatment combinations, he and his colleagues slashed mortality rates to 50%, reversing the negative perception of quarantine and thus helping to stem the contagion.

By setting aside your preconceptions, you become a sharper observer of what people say and do. This changes not just *how* you pay attention but also *whom* you pay attention to—and previously unconsidered niche populations often reveal unsuspected pain points. The toy group Lego learns a lot from the frustrations of its adult enthusiasts,

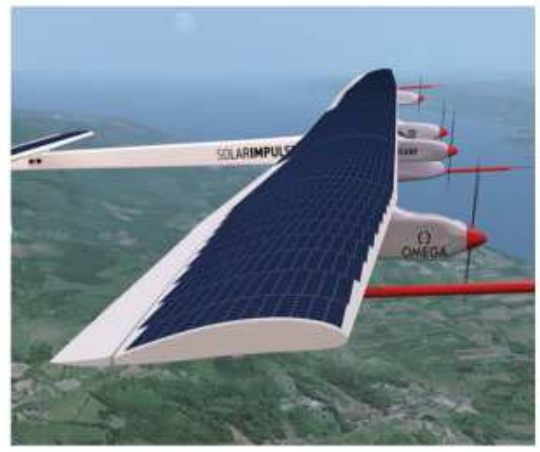
the cleaning-products giant S.C. Johnson from observing hygiene-obsessed OCD sufferers, and IKEA from trying to understand what "IKEA hackers," who customize and repurpose the furniture maker's goods, are "trying to tell us about our own products."

Digital technologies allow the tracking of behavior on a much larger scale than was previously possible, offering complementary ways of detecting tacit needs. In health care, for instance, researchers are studying the lived experience of Parkinson's disease by having volunteers use their smartphones to measure tremors (thanks to the function that captures portrait and landscape views), muscle tone (the microphone indicates the strength of the patient's voice box), involuntary movements (the touch screen records them), and gait (if the phone is in a pocket, it senses the patient's unsteadiness). The researchers can thus track the efficacy of medication not just before and after dosing but over time. And they get a rich picture of what participants actually do, as opposed to what they say they do.

Cyberspace can also help companies identify expert users in their practice communities. Medical device companies could glean insights from the online forums of "body hackers"—people who implant microchips, magnets, LED lights, and other technology in themselves with the aim of augmenting human capabilities. Inspired by that ethos, Medtronic is considering how its pill-sized pacemaker could be enhanced and implanted in healthy people to give them biometric feedback and improve their lifelong care.

Companies can use digital technology to engage with trendsetters directly or eavesdrop on user forums and blogs for clues about evolving needs. In 2009 Nivea conducted an online analysis, or "netnography," of discussions about deodorant use across 200 social media sites. Contrary to expectations, the key preoccupation was not fragrance, effectiveness, or irritation but the staining of clothes. This insight paved the way for a new category of antistain deodorants in 2011, the most successful launch in the company's 130-year history. In the public sector, online media analysis is being used to explore issues such as exercise, generic drugs, and—in an effort to improve health care social workers' interventions—resistance to vaccination.

Digital technologies can't replace direct observation, of course. But they expand the number and type of insights



A 3-D model of Bertrand Piccard's solar-powered plane in 2007

The C-shaped blades developed by Van Phillips to help amputees run and jump

"Shrek models," hand-built by Frank Gehry to learn from clients' discomfort

The first digital camera, created by Steven Sasson in 1973 while he was working for Eastman Kodak

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How Robust Is Your Creative Process?

To assess your capacity to innovate, answer the following questions and calculate your average score for each element. Scores below 4 indicate areas for improvement.

	Disagree → Agree						
Attention							
I periodically try to experience what it feels like to be a user/customer.	1	2	3	4	5	6	7
I notice subtle developments or surprising things that others have overlooked.	1	2	3	4	5	6	7
I use digital tools to constantly monitor my environment.	1	2	3	4	5	6	7
I am curious about customers who behave in very different ways.	1	2	3	4	5	6	7
Perspective							
I make time to consider the big picture, even when I am deep in details.	1	2	3	4	5	6	7
I attend events that have nothing to do with my core industry.	1	2	3	4	5	6	7
I try to challenge my default perspective on problems.	1	2	3	4	5	6	7
I seek out people with a different view on the issues I am looking at.	1	2	3	4	5	6	7
Imagination							
I openly question accepted practices and assumptions.	1	2	3	4	5	6	7
I mix ideas from unrelated areas.	1	2	3	4	5	6	7
I try to be creative in how I make connections between things I observe.	1	2	3	4	5	6	7
Experimentation							
I try to find novel ways of demonstrating the value of my ideas.	1	2	3	4	5	6	7
I use computer simulations to test the feasibility of my ideas.	1	2	3	4	5	6	7
I often don't settle on the first solution I come up with, or the second.	1	2	3	4	5	6	7
I allow myself to fail as long as I learn from it.	1	2	3	4	5	6	7
Navigation							
I have access to people who can help champion my solution.	1	2	3	4	5	6	7
I fully understand the varying interests of critical stakeholders.	1	2	3	4	5	6	7
I know how to tailor and deliver a message for different audiences.	1	2	3	4	5	6	7
I have a good sense of when to disclose strategic information.	1	2	3	4	5	6	7

generated, providing access to a wealth of unfiltered and unstructured user-generated content that people can then make sense of.

Perspective: Step Back to Expand Your Understanding

Having zoomed in to gather insights about a situation, a need, or a challenge, you must then pull out to gain perspective, fighting against framing and action biases that might encourage you to accept the issue as presented and rush into problem solving.

To process what you have learned, detach: Change activities, or take a strategic break. During his third attempt to circle the globe by balloon, in 1999, the Swiss psychiatrist and adventurer Bertrand Piccard was obsessed by fuel conservation. After completing the exploit, with barely any liquid gas to spare, he realized that he had spent his 20 days aloft in constant fear of running out. As he waited (for half a day) to be picked up from the Egyptian desert, it dawned on him that the core problem was not how to manage fuel but how to manage *without* fuel. Redefining the issue in this way set the stage for his next circumnavigation challenge—in a fully solar-powered plane.

His insight occurred only after Piccard stepped back. It's not easy to prime yourself for inspiration while you're in the thick of action. Consider the pioneering chef Ferran Adrià, who melded haute cuisine, art, and science; generated more than 1,800 signature dishes over 20 years; and earned his restaurant, El Bulli, the rating of "world's best" a record five times. The key to his creativity, Adrià once explained to HBR, was closing his restaurant for six months each year. "The pressure to serve every day doesn't offer the kind of tranquility necessary to create as we would like," he said. "The most important thing is to leave time for regeneration." This mindset is reflected in the Japanese concept *ma,* which stresses that space is necessary for growth and enlightenment.

Digital tools can help create that space, freeing up your time through automation and increasing your capacity to pause and make sense of weak signals. Take the experience of Nancy Lublin. While serving as CEO of DoSomething, a

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global nonprofit that works with young people to effect social change, she ran charity drives that mobilized volunteers by text. Along with messages from people looking to donate their time—some 200,000 messages in a typical campaign—incoming texts would include a handful of unrelated messages from distressed teenagers. Staffers would respond with referrals to relevant helplines, until one particularly disturbing message, from a girl being raped by her father, forced them to rethink their approach.

After two weeks of sleepless nights, Lublin realized why adolescents were contacting her organization with problems unrelated to its cause: Texting is anonymous, private, and quiet. The cries for help pointed to an unmet need, leading Lublin to create Crisis Text Line (CTL), a free service that offers round-the-clock counseling and intervention. Although launched without fanfare, it has outpaced even Facebook's reach throughout the United States.

Imagination: Look for Unexpected Combinations

To produce a truly original idea, you must free your imagination, challenge orthodoxy, and envision that which is not. But "functional fixedness" often limits the ability to think creatively or to conceive of alternative uses for familiar objects and concepts. Overcoming this barrier requires asking uninhibited questions such as "Why not?" and "What if?"—as Van Phillips did when he queried the requirement that prosthetic limbs resemble human ones. His now-familiar C-shaped blades help amputees run and jump much the way able-bodied athletes do.

To spur imagination, organizations may ask questions such as "What if we no longer did what we do now?"—not necessarily because they intend to abandon current activities but as a way to envision connections between existing strengths and new opportunities. In 2009 the McLaren Group, known for Formula 1 racing, asked precisely that question—which liberated it to think about how its capabilities in materials science, aerodynamics, simulation, predictive analytics, and teamwork might apply to other sectors. It leveraged those capabilities to improve the performance of clients ranging from elite rowing, cycling, and sailing teams to health care systems and air traffic control services. As a result, it has morphed into a consulting and technology group that happens to have a successful Formula 1 team.

Imagination is often seen as something mystical and inaccessible, but it is chiefly a matter of positing unexpected combinations. At the most basic level, you can apply an existing solution from one domain to another. Jonathan Ledgard, a war reporter and a former longtime Africa correspondent for the *Economist,* imagined a drone-based network for delivering medical supplies to remote areas in Africa. Thus was born Redline, which began trial operations in Rwanda in 2016. Because the price and efficiency of drone batteries has not kept pace with advances in airframes, Ledgard has begun to focus on developing droneports to support the air cargo routes.

Outsiders often find it easier than insiders do to connect disparate thoughts, because they come to the table with fewer preconceptions. Phillips was not an engineer and had never worked in a prosthetics lab; his running blades were inspired by his experience as a pole vaulter and a springboard diver prior to losing a leg in an accident. The latest medical innovation for assisting difficult births—successfully tested in South America and recently licensed by Becton, Dickinson—was the brainchild of an Argentinian car mechanic, Jorge Odón. Enthralled by a YouTube video on using a plastic bag to retrieve a cork from inside an empty wine bottle, the father of five realized that the same principle could save a baby stuck in the birth canal. As a chief medic at the World Health Organization told a *New York Times* reporter, "An obstetrician would have tried to improve the forceps or the vacuum extractor, but obstructed labor needed a mechanic."

Research confirms that distance often helps in coming up with novel ideas. In one study, separate groups of carpenters, roofers, and in-line skaters were asked for ideas on how to improve the design of carpenters' respirator masks, roofers' safety belts, and skaters' kneepads. Each group was much better at coming up with solutions for the fields outside its own.

Organizations try to spark such connections by bringing together people with diverse knowledge bases and perspectives. Technology allows them to tap the wisdom of experts far beyond their networks—people who may have no familiarity with a given market or industry but can bring different experiences to bear. For example, in 2007 the Oil Spill Recovery Institute used crowdsourcing to tackle the long-standing problem of cleaning up oil spills in subarctic waters. The

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winning idea came not from an oil industry specialist but from a chemist, John Davis, who applied his expertise in the concrete industry to devise a means of keeping oil liquid as it is pumped. Examining 166 problem-solving contests posted on the InnoCentive innovation platform, Harvard Business School's Karim Lakhani found that winning entries were more likely to come from "unexpected contributors" with "distant fields" of expertise.

In addition, the digital revolution has blurred industry boundaries, facilitating unexpected combinations. The data routinely collected in one area can benefit players in unrelated fields, opening up new uses for underleveraged assets. Melinda Rolfs, an executive at Mastercard, which sells its anonymized transaction data and analytics to merchants and financial institutions, realized that the data could be made available free to charities and other nonprofits to bolster their fundraising efforts. She now spearheads Mastercard's data philanthropy program, creating value in new ways for the business and for society.

Experimentation: Test Smarter to Learn Faster

Experimentation is the process of turning a promising idea into a workable solution that addresses a real need. The big risk, once you start testing, is that confirmation biases and sunk cost effects will deaden your responsiveness to corrective feedback. Successful innovators design their experiments to learn faster and cheaper, and they remain open to sharp changes in direction. They test to *improve* rather than to *prove*.

The lean start-up methodology puts learning at the heart of its approach, and although that objective is certainly important, it often conflicts with the emphasis the model's proponents also put on speed. Frenetic cycles of build-measure-learn encourage organizations to settle for a "good enough" product-market fit, leading them to miss more-ambitious solutions.

Instead, externalize your idea early and often so that others can visualize, touch, or interact with it. When pitching his idea to doctors, Jorge Odón used a glass jar for a womb, a child's doll for a trapped baby, and a fabric bag sewn by his wife as his lifesaving device. This kind of low-cost mock-up is sometimes dubbed a Frankenstein prototype.

Negative reactions are as valuable as positive ones and are critical to avoiding costly errors. To take an extreme example, the pioneering architect Frank Gehry produces what he calls Shrek models, intended to make clients uncomfortable. ("Shrek" is Yiddish for "fear.") Successive models don't build on the ones before them; rather, they depict divergent approaches, enabling Gehry to explore and learn from client discomfort and allowing the ideas to mature.

Be careful not to overinvest in your prototype. The aim is to test without building. One way is to use off-the-shelf technology or human intervention to fake a functional product or service—a so-called Wizard of Oz prototype. Researchers working on robot-assisted therapy for health care interventions typically assess user reactions to proposed designs with a Wizard of Oz technique: Without the patient's knowledge, the "robot" is controlled by a human operator. This kind of mock-up lets the researchers explore their concepts before writing any code.

Digital tools are a great aid to simulation. A website or a video can create the illusion of an offering before it exists—as Dropbox did when it made a video demo of the prototype for its file-sharing software to avoid bringing to market a product no one would want. Digital technology also simplifies A/B testing, whereby you propose two versions of an offering to learn what users value most. Nancy Lublin initially planned to use counselors from paid crisis centers on the hotline but changed her mind after comparing their performance with that of volunteers trained in best practices that had been identified by digital text analysis. One salient finding from the analysis: "I" statements—discouraged in traditional counseling—were three times as effective as other statements in keeping discussions going. Armed with this type of knowledge, the volunteers outperformed their professional counterparts on every key indicator: They were faster and cheaper and got higher-quality ratings. This persuaded Lublin to flip her business model, ditching the paid counselors in favor of trained volunteers.

Digital advances help you get closer to the ultimate goal in testing: trial without real-world error. To build his solar-powered plane, Bertrand Piccard approached conventional aircraft manufacturers, but they showed little interest. So he found partners outside the industry—80 in all. In conjunction with an advanced software and solutions provider, they

created a "digital twin" of the plane, using 3-D software to design and test the individual parts and complex assemblies. That allowed them to forgo costly and slow physical prototypes and simulate the plane's performance under a variety of conditions, dramatically reducing the number of dead ends. What started as a purely technological experiment became an experiment in virtual collaboration as well.

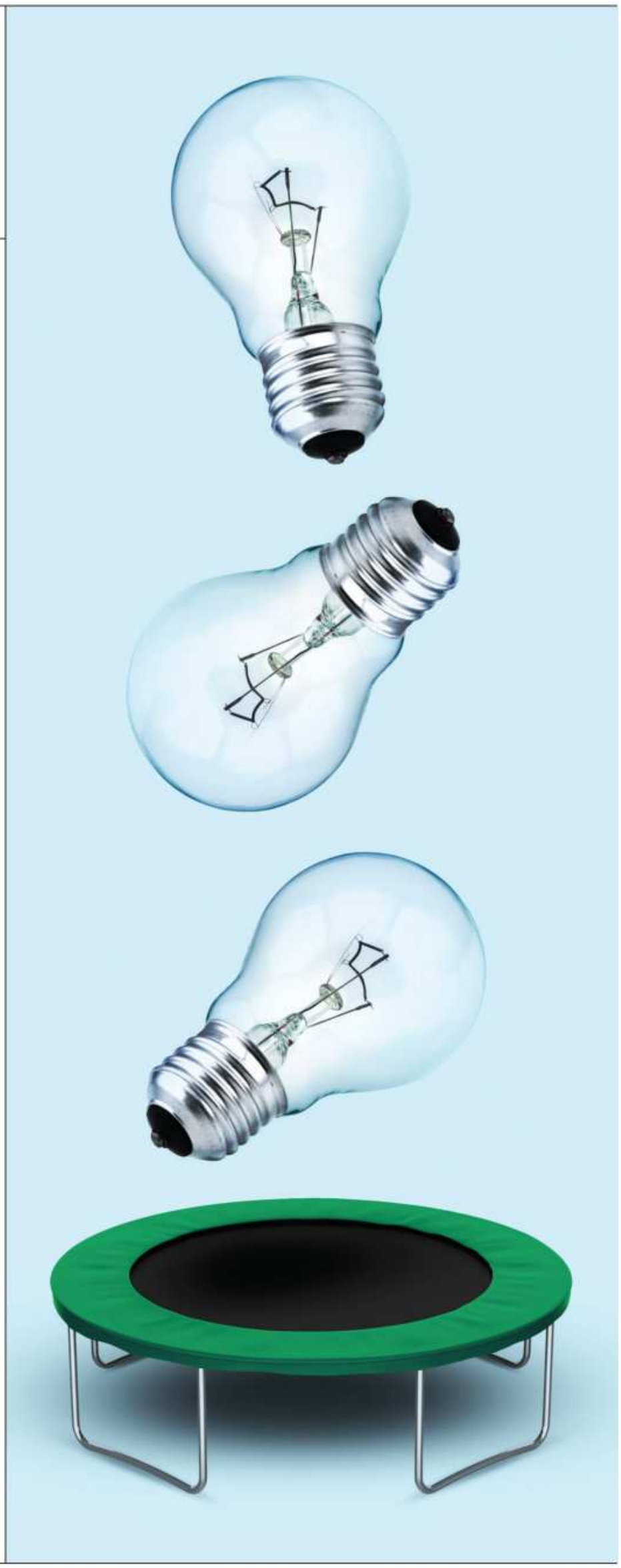
Navigation: Maneuver to Avoid Being Shot Down

To bring your idea to fruition, you'll need to adjust to the forces that can make or break it. But your belief in your idea—and your overfamiliarity with the context—may lead you to underestimate the effort needed to mobilize supporters and steer past obstacles. You must sharpen your reading of hostile environments, including your organization's own immune system, and meet multiple persuasion challenges. Original thinking is essential in shaping your business model as well as your offering.

The way an idea is framed affects how people perceive its value. Steve Sasson, the Kodak engineer who invented the digital camera, acknowledges that dubbing his innovation "filmless photography" was a serious hindrance to gaining internal support at a company whose very existence revolved around film. Your enthusiasm may blind you to the threat you pose to others, and your offering—often developed under the radar, with input only from trusted critics—may not survive first contact with skeptics.

By contrast, when Jonathan Ledgard conceived of drones as an answer to Africa's transport challenges, his adroit framing paved the way for acceptance: He called the unmanned aircraft "flying donkeys" (a reference to the loads they could carry), rendering a potentially menacing concept unthreatening, concrete, and sticky. An engaging frame uses the familiar to explain the less familiar and emphasizes continuity. If it resonates, digital channels can accelerate interest, as they did for Narayana Peesapaty's edible cutlery.

Presenting a disruptive innovation in a way that is responsive to the collective DNA of your organization is also crucial. According to Jean-Paul Bailly, who oversaw a dramatic transformation of the French National Mail Service (La Poste) from 2002 to 2013, "You have to demonstrate that the change



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can help you remain true to your identity." In 2010, even as the state-owned entity began to privatize, Bailly continued to emphasize public service values and public trust. Those core assets underpinned new activities, including e-commerce, banking, and mobile telephony. Building on public confidence in mail carriers, the organization added services for the elderly—an innovative response to both the country's aging population and the declining number of letters sent by post. Customers can now commission the local mail carrier to drop in on aging relatives, and postal workers are trained in the situations they might encounter when they do. Because of such measures, the transformation was completed without any layoffs, and the mail service's revenues have continually grown.

Even if you've managed to secure internal buy-in and user interest, remember that an entire ecosystem stands between you and those you hope to serve. The input, connections, or cooperation of an array of stakeholders could determine the outcome of your offering. Owlet devised a wireless wristband to monitor hospital patients' vital signs. The development team thought it had a winner, because neither patients nor nurses liked the wired products. But that issue was not a pain point for hospital administrators, who refused to pay for the wristbands. In focusing solely on users, Owlet had neglected buyers. It eventually hit the mark with a smart sock that tracks a sleeping infant's pulse and breathing and, when warranted, sends an alarm to the caregiver's smartphone. This time, crucially, the buyers—anxious parents—were also beneficiaries.

Successful navigation isn't just a matter of anticipating blockers; gaining support from unconventional allies can also be essential. The Mexico-born theme park KidZania is an indoor "city" where children can role-play adult jobs. When its founders ran out of development funds, they approached corporate sponsors—not just for financial support but also for professional expertise, facilitating a more realistic experience in terms of props, activities, architecture, and interiors. KidZania's industry partners create mini versions of their stores, banks, and offices, adding authenticity to the role-playing: Children can deliver packages dressed as DHL drivers or train to be pilots on a British Airways flight simulator. The concept took off, and KidZania became the world's fastest-growing group in experiential learning for children, with operations on five continents.

Digital technologies offer further opportunities for novel collaborations. For example, Vestergaard Frandsen (VF), a Switzerland-based disease-control company, devised an ingenious line of water filters, but they were too costly for the rural African and Indian communities that needed them most. Because the filters reduce emissions by eliminating the need to purify water over open fires, VF came up with the idea of distributing the devices free through funding from carbon offsets. To unlock that novel source of funding, it had to satisfy independent auditors that hundreds of thousands of filters were indeed being used. The platform's developers drew on an open-source data-collection platform developed at the University of Washington to create a smartphone app that would let field representatives photograph recipients of the filters and record their homes' GPS coordinates. Thus each recipient was reachable for follow-up and auditing purposes—making the solution both scalable and sustainable.

A Flexible Sequence

For the sake of convenience, we've presented our framework as a kind of process. In practice, though, the five elements constitute not an orderly sequence or even a cycle but a mix that involves frequent crisscrossing among activities. This accounts for two realities that are often overlooked by conventional innovation methodologies:

Multiple entry points. Although *attention* is a logical starting point for innovation, others are valid too. *Imagination* is a common gateway. Jorge Odón was not looking to improve on

FURTHER READING

ATTENTION

"Becoming a First-Class Noticer"
Max H. Bazerman
HBR, July–August 2014

A Practical Guide for Improving Your Observational Skills
James H. Gilmore
Greenleaf Book Group, 2016

An entire ecosystem stands between you and those you hope to serve. The input, connections, or cooperation of an array of stakeholders could determine the outcome of your offering.

existing birthing equipment; the idea literally came to him in his sleep. Imagination was also the entry point for Bertrand Piccard's solar plane. His story shows that to achieve a breakthrough innovation, you don't necessarily have to know something no one else does; you can get there by believing in something no one else believes in. Design thinking has difficulty accommodating such big-leap innovation, which is based largely on a top-down belief in possibility rather than on present needs or technologies. Part of our genius as human beings is imagining that which is currently out of reach. Still-immature technology that might be available in a few years is not a direct focus of design-thinking methodologies.

Another entry point is *experimentation,* as when you stumble upon a finding that leads you not to merely pivot but to reboot. A few years ago Jeannette Garcia, a chemist at IBM Research who was seeking to synthesize a particular polymer, set up a chemical reaction and stepped away to fetch an ingredient. Returning to the flask, she found a bone-hard substance: It turned out she had discovered the first new class of polymers in decades. The new substance was superstrong, lightweight, and, unlike comparable materials, easily and infinitely recyclable. That unique combination makes it a breakthrough discovery with a wide range of potential applications in aerospace, autos, electronics, and 3-D printing, although for now it remains a solution in search of meaningful problems.

Multiple pathways. Just as you can start anywhere in your creative process, you can proceed in any direction and switch focus as required. Existing innovation models do not explicitly acknowledge such freedom—and so they are often taken too literally and reduced to rigid, unrealistic recipes.

Consider Garcia's polymer discovery. Looking ahead, IBM has twin priorities to juggle: It must define relevant customer needs (attention) while deciding with whom it should partner (navigation) to bring a solution to fruition. This may involve much back-and-forth along with further experimentation (to produce prototypes) and occasionally stepping back (to maintain perspective and avoid overstretching).

The creative process is full of dead ends that may require you to revisit your original question, the options for responding, or your choice of partners (as CTL did when it pivoted away from established crisis professionals). Insights may occur after a partnership has been formed. Knowledge gained through experimentation may deepen your understanding of users and the problems they face. Sometimes you will address the challenge of asking the right people the right questions late in the process—as when Owlet realized that wireless health monitors were a pressing need not for hospital administrators but for new parents. Follow whatever sequence works for you. Process fixedness is itself a barrier to breakthrough innovations.

A caveat in closing: Although the order is flexible, you do need to touch all the elements at least once, because each one neutralizes different biases. Neglecting even one can lead you to focus on the wrong problem, idea, or solution. Creativity is a journey of sense making. By attending to all five elements, you will maximize your chances of reaching a truly game-changing innovation at the journey's end.

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